



UNDERWATER ROBOTICS SYSTEM ARCHITECTURE

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COMPONENTS



- Hull/Body: Pressure-resistant casing containing internal components.
- Propulsion: Electric motors powering thrusters or biomimetic fins.
- Control System: Embedded computer handling navigation, data processing, and communication.
- Sensor Suite: Includes sonar, IMU (Inertial Measurement Unit), depth sensors, and cameras.
- Power Supply: Battery (usually lithium-based), possibly rechargeable via tether or docking station.

COMMUNICATION



Acoustic Communication:

- Most common method underwater (radio and GPS do not work well underwater).
- Slow data rates ($\sim$ kbit/s) with high latency.

Tethered Communication:

- Used in ROVs, allowing real-time control and data transfer via a cable.
- Optical Communication (experimental):
- High data rate, but limited range and susceptible to turbidity.

GROUND CONTROL STATION



- ROVs: Controlled in real time from the surface via joystick or software.
 - AUVs: Mission parameters are uploaded in advance, and post-mission data is downloaded upon retrieval.
 - Interfaces:
 - Real-time telemetry
 - Map overlays
 - Live video and sensor feeds
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THANK YOU

